# CSE 270 Sports Analytics

Homework 2

Name: _____

## Poisson Regression

### Problem 1(20 points)

**Predicting football games using Poisson regression.**

1. Make predictions for the same game from the same league assigned to you in homework 1, but now use the poisson regression model instead of distributions.**(8 points)**
2. Interpret the result and compare with the betting odds from homework 1.
   Is the same team expected to win ? **(2 points)**
3. Define a metric to calculate the efficiency of the predictions. **(5 points)**
4. Find the under estimated and over estimated teams based on Poission model. **(5 points)**

## Bradley terry model

### Problem 2(30 points)

**Construct Bradley-Terry model for NBA regular season games.**

1. Load the dataset **nba_2009_2021** from the package **SportsAnalytics270** and remove the game Boston Celtics vs Indiana Pacers by filtering the **home.PTS** and **away.PTS** variables to be equal to 0.**(3 points)**
2. Prepare the dataset for fitting a Bradley-Terry model by adding 2 variables, **ht_w** and **at_w**.
   Assign the values of the varibles to (**ht_w** = 1, **at_w** = 0) if the home team won the game and (**ht_w** = 0, **at_w** = 1) if the away team won the game. **(5 points)**
3. Convert the variables representing the home team and the away team into a factor. **(2 points)**
4. Fit a Bradley-Terry model into the dataset to get the abilities of the teams to win. **(5 points)**
5. Plot the abilities of the teams. **(5 points)**
6. Interpret the results of the plot, which team is the best and which team is the worst ? Why is the ability to win for team **Atlanta Howks** equal to 0 ? **(5 points)**
7. Make a prediction for the upcoming 3 games.
   (Schedule can be found here: https://www.nba.com/schedule) **(5 points)**

# Winning Percentage

## Pythagorean wins for european football

## Problem 3 (30 points)

1. Library **SportsAnalytics270** has a function final_table. It creates final league standing for the season. Use this function to create a dataframe with final standings of all seasons of your league. Combine seasons into 1 datafrmae. You need to get something like nba_east. **(10 points)**
2. Create a variable for winning percentage and goal difference. Note, as there are draws in football, we are going to take draw as a half win. **(2 points)**
3. Plot Goal differential against Winning Percentage with a regression line.
   Make sure to have a title and apropriate axis labels for your plot. **(5 points)**
4. Interpret the plot, do you think there is a strong correlation between the variables ? **(3 points)**
5. Calculate Pearson correlation coefficient for Goal differential and Winning Percentage. **(2 points)**
6. Build a regression model to estimate the value for k from the Pythagorean formula **(3 points)**
7. Using the estimated value for k, calculate Pythagorean Winning percentage and Pythagorean wins for each team. **(3 points)**
8. Find over performing and under performing teams. **(2 points)**

# Elo Ratings

## Problem 4 (20 points)

**Building Elo rating model for NBA games.**

1. Calculate the relative Elo ratings for all the games in the nba dataset.
   Make sure to adjust the ratings of the teams at the beginning of each season.
   This link https://fivethirtyeight.com/features/how-we-calculate-nba-elo-ratings/ shows the steps for calculating elo ratings in nba. **(10 points)**
2. Interpret the predictive power of your model using either confusion matrix or brier score. **(5 points)**
3. Make predictions for the same games you chose in **Problem 2.7**, but now use Elo rating model.
   **(3 points)**
4. Compare the results **(2 points)**
5. Try to play with the home advantage parameter, K value or anything else, either on football data or in nba and check if that will help to improve your models' predictive power.
   **(Bonus question 10 points)**